

Spenser Millburn – Resume

Full Stack Software Engineer

Professional Summary

Full Stack Engineer with expertise in both frontend and backend development, microservices architecture, and cloud deployment. Proven ability to architect and implement end-to-end solutions that deliver exceptional user experiences while meeting complex technical requirements. Excels in cross-functional team environments.

Technical Skills

Frontend: TypeScript, React, WebSockets, HTML/CSS, UI/UX Design, Web Application Architecture

Backend: Python, Go, Flask, C++, Node.js, RESTful APIs, Microservices, Linux Systems, Bash

DevOps/Cloud: Docker, Azure, ETL Data Pipelines, CI/CD, Systemd, Cloud Deployment

Database: SQL, NoSQL, Data Acquisition & Analysis

Education

Northeastern University: Mechatronics Engineering, B.S.

Class of 2025, GPA 3.7

Professional Experience

(September 2022 – Current) – Senior Software Engineer - *Walmart Advanced Systems & Robotics* → *Symbolic*

- Designed and implemented ETL data pipelines that processed mission-critical telemetry data, deployed to Azure cloud environment.
- Containerized development toolchain with Docker, improving build consistency and deployment reliability.
- Developed hierarchical state machines using event-driven architecture for orchestration of complex systems.
- Created high-performance, real-time data collection and analytics infrastructure to improve observability stack.
- Mentored junior engineers and interns on software best practices and new technologies.

(March 2022 – September 2022) – Full Stack Engineer – *Building Machines Incorporated*

- Designed, implemented and tested full stack application with React frontend and Flask Python backend.
- Built responsive UI that replaced physical control panel, enabling remote monitoring and control of industrial systems.
- Developed WebSocket interface for real-time communication between frontend and backend components.
- Created and deployed Linux services using Systemd for reliable application management.
- Implemented data acquisition, analysis and visualization tools with JupyterLab and Python.

(March 2021 – March 2022) – Software Development Engineer – *Owl Labs*

- Led customer-facing Embedded Linux and Android software development initiatives.
- Developed data acquisition and visualization tools for analyzing thermal, acoustic, and airflow systems.
- Created dashboards and analytics tools using Python, NumPy, and Matplotlib.
- Supported 30+ engineers with software tools to improve development workflows.
- Worked cross-functionally with hardware, testing, and product teams to ship consumer products.

(August 2020 – March 2021) – Software Engineer – *Degree Controls Systems*

- Designed and implemented customer-facing application with Python and PyQt5 GUI framework.
- Created intuitive user interfaces for complex engineering systems.
- Developed software for hardware testing and validation processes.
- Managed engineering change order documentation and implementation.
- Collaborated with cross-functional teams to deliver integrated hardware-software solutions.

(August 2018 – August 2020) – Software Developer – *Sartorius NA*

- Architected and built custom IoT automation server framework in Python, generating over \$50,000 in recovered value.
- Developed web-based GUI for dynamic project management and reporting.
- Created microcontroller clients reporting live production data via socket server architecture.

- Automated paperwork systems, targeting KPI calculations and metrics.
- Implemented automated label printing and generation system.

Leadership & Team Experience

- **Project Manager & Software Lead** for NEU Robotics Quadraped Project, coordinating 75+ engineering students.
- Led cross-functional software team developing architecture for complex robotic platform.
- Managed and coordinated 10+ subteams across various technical domains.
- Developed and documented software architecture standards and workflows.
- Implemented agile methodologies to improve team productivity and project outcomes.